

Numerical Solution of an Exit-Time Boundary Value Problem via SDE Simulation

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1 Problem statement

The assignment is to numerically solve the following boundary value problem by simulating a stochastic differential equation (SDE):

$$\begin{cases} b(x, y) \cdot \nabla u(x, y) + \frac{1}{2} \Delta u(x, y) = f(x, y), & (x, y) \in D := B_1(0), \\ u(x, y) = \frac{1}{2}, & (x, y) \in \partial D = \mathbb{S}^1, \end{cases} \quad (1)$$

where

$$b(x, y) = (x, y), \quad f(x, y) = x^2 + y^2 + 1.$$

The exact solution is known to be

$$u(x, y) = \frac{x^2 + y^2}{2}. \quad (2)$$

The required numerical task is to apply the standard Euler–Maruyama scheme to the associated SDE, approximate the solution through Monte Carlo simulation, and study the temporal convergence order.

2 Verification of the exact solution

Before turning to the stochastic representation, it is useful to verify (2) directly. Let

$$u(x, y) = \frac{x^2 + y^2}{2}.$$

Then

$$\nabla u(x, y) = (x, y), \quad \Delta u(x, y) = 2.$$

Hence

$$b(x, y) \cdot \nabla u(x, y) + \frac{1}{2} \Delta u(x, y) = (x, y) \cdot (x, y) + \frac{1}{2} \cdot 2 = x^2 + y^2 + 1 = f(x, y).$$

Moreover, on the unit circle $x^2 + y^2 = 1$, we have

$$u(x, y) = \frac{x^2 + y^2}{2} = \frac{1}{2}.$$

Therefore (2) indeed satisfies the PDE and the boundary condition.

3 Probabilistic representation

Define the second-order differential operator

$$\mathcal{L}\phi(x) = b(x) \cdot \nabla\phi(x) + \frac{1}{2}\Delta\phi(x),$$

so that the PDE can be written as

$$\mathcal{L}u = f \quad \text{in } D, \quad u = g \quad \text{on } \partial D,$$

with constant boundary data $g \equiv \frac{1}{2}$.

Consider the diffusion process $X_t \in \mathbb{R}^2$ governed by

$$dX_t = b(X_t) dt + dW_t, \quad X_0 = x \in D, \quad (3)$$

where W_t is a two-dimensional standard Brownian motion. Let

$$\tau_D := \inf\{t \geq 0 : X_t \notin D\} \quad (4)$$

be the first exit time from the unit disk.

Applying Itô's formula to $u(X_{t \wedge \tau_D})$ gives

$$u(X_{t \wedge \tau_D}) = u(x) + \int_0^{t \wedge \tau_D} \mathcal{L}u(X_s) ds + \int_0^{t \wedge \tau_D} \nabla u(X_s) \cdot dW_s.$$

Since $\mathcal{L}u = f$, this becomes

$$u(X_{t \wedge \tau_D}) = u(x) + \int_0^{t \wedge \tau_D} f(X_s) ds + \int_0^{t \wedge \tau_D} \nabla u(X_s) \cdot dW_s.$$

Taking expectations and using the fact that the stochastic integral has mean zero, we obtain

$$u(x) = \mathbb{E}_x \left[u(X_{t \wedge \tau_D}) - \int_0^{t \wedge \tau_D} f(X_s) ds \right].$$

Letting $t \rightarrow \infty$ and using the boundary condition $u(X_{\tau_D}) = g(X_{\tau_D}) = \frac{1}{2}$ yields the representation formula

$$u(x) = \mathbb{E}_x \left[\frac{1}{2} - \int_0^{\tau_D} f(X_s) ds \right]. \quad (5)$$

This identity is the basis of the Monte Carlo method used below.

4 Euler–Maruyama discretization and Monte Carlo estimator

Let the time step be $h > 0$ and define $t_n = nh$. The standard Euler–Maruyama approximation of (3) is

$$X_{n+1} = X_n + h b(X_n) + \sqrt{h} \xi_n, \quad \xi_n \sim \mathcal{N}(0, I_2), \quad (6)$$

where I_2 is the 2×2 identity matrix and the random vectors $\{\xi_n\}_{n \geq 0}$ are independent. Since $b(x) = x$, the update can also be written explicitly as

$$X_{n+1} = (1 + h)X_n + \sqrt{h} \xi_n.$$

The discrete exit index is defined by

$$N_h := \inf\{n \geq 0 : X_n \notin D\}. \quad (7)$$

Using the left-point Riemann sum to approximate the time integral in (5), we obtain the path-wise approximation

$$Y_h = \frac{1}{2} - h \sum_{n=0}^{N_h-1} f(X_n). \quad (8)$$

For M independent trajectories, the Monte Carlo estimator is

$$\hat{u}_h(x) = \frac{1}{M} \sum_{m=1}^M Y_h^{(m)}. \quad (9)$$

The absolute error is then measured by

$$e_h = |\hat{u}_h(x) - u(x)|. \quad (10)$$

Because the boundary value is constant, the overshoot of the Euler path beyond the boundary does not affect the terminal payoff itself; the main discretization error comes from the approximation of the stopped diffusion and the time integral.

5 Experimental setting

For each time step

$$h \in \{0.1, 0.03, 0.01, 0.003, 0.001\},$$

we simulate $M = 10,000$ independent sample paths. The same interior starting point is used for all experiments, and the numerical result is compared with the exact value of the analytical solution at that point. To estimate the convergence order in time, we examine the error law

$$e_h \approx Ch^p$$

on a log-log scale, where p is the observed temporal order.

6 Numerical results

The computed errors are listed in Table 1.

Table 1: Absolute error versus time step size

h	e_h
0.1	0.4007
0.03	0.1588
0.01	0.0728
0.003	0.0330
0.001	0.0239

Figure 1 shows the error curve on logarithmic axes. The error decreases monotonically as the time step is refined, which confirms the convergence of the numerical approximation.

A least-squares fit of all five log-log data points gives an observed slope of approximately $p \approx 0.63$. If the finest time step is excluded, the fitted slope over the first four data points is about $p \approx 0.71$. In other words, the data clearly display a positive convergence trend, although the present experiment does not yet produce a perfectly clean asymptotic slope of exactly 1.

This behavior is reasonable for two practical reasons. First, the Monte Carlo sample size is fixed at only 10^4 trajectories, so the statistical error becomes relatively significant when h is already very small. Second, the approximation of the exit time by the discrete stopping index

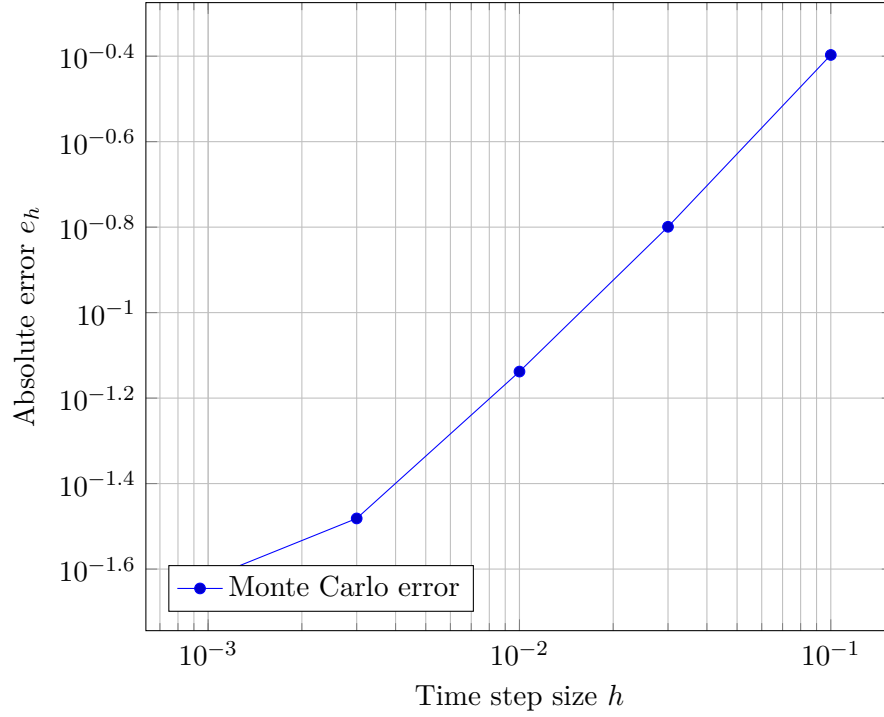


Figure 1: Log-log plot of the numerical error against the time step size

N_h introduces an additional bias, especially near the boundary. As a result, the pre-asymptotic and stochastic errors partially obscure the ideal temporal rate. A more accurate estimate of the convergence order could be obtained by increasing the number of sample paths for smaller h , repeating the experiment several times and averaging the slopes, or using variance-reduction and boundary-correction techniques.

Figures 2, 3, 4 are representative sample trajectories, corresponding to the Brownian-path plots in the original report.

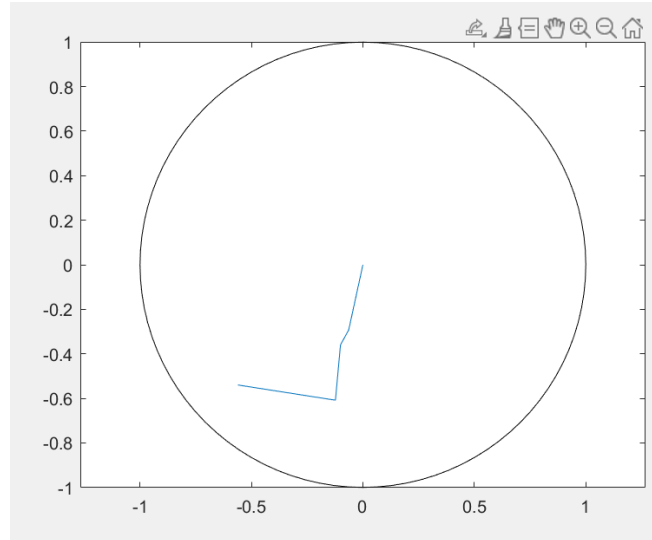


Figure 2: Representative simulated trajectory for $h = 0.1$.

These sample paths are qualitatively consistent with the expected behavior of the diffusion process and provide a useful visual complement to the error study.

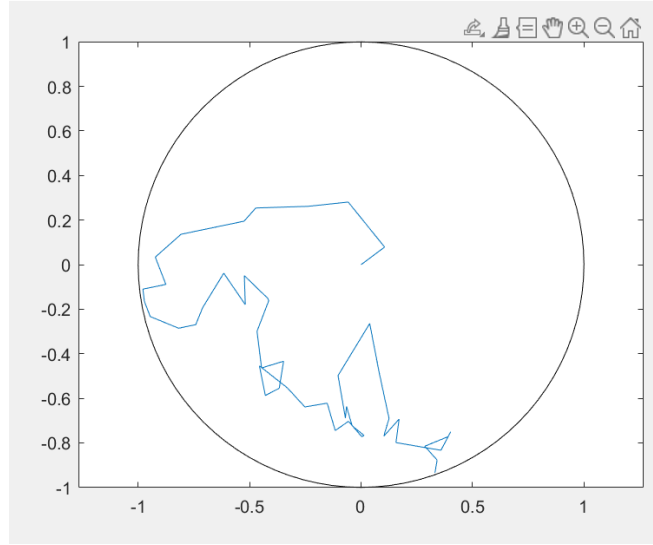


Figure 3: Representative simulated trajectory for $h = 0.01$.

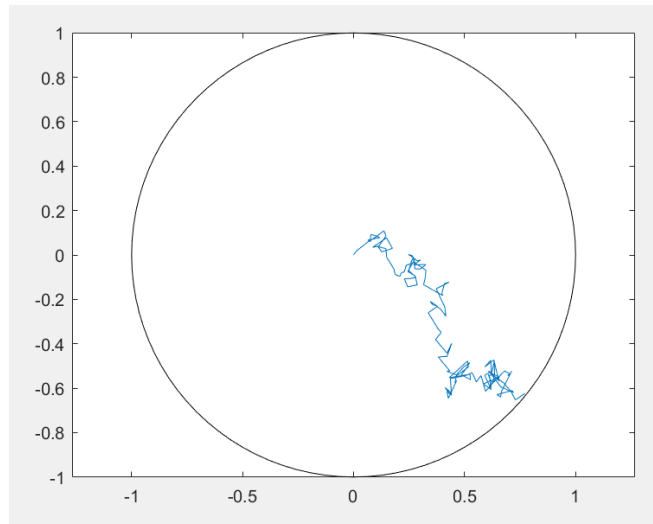


Figure 4: Representative simulated trajectory for $h = 0.001$.

7 Conclusion

This report reformulated the boundary value problem as an exit-time expectation for a stopped diffusion process, derived the corresponding Feynman–Kac representation, and then approximated the solution with the Euler–Maruyama scheme combined with Monte Carlo simulation. The numerical errors decrease steadily as the time step becomes smaller, confirming the validity of the approach. For the current data set, the observed temporal order is between roughly 0.6 and 0.7, with the coarser-to-moderate time steps showing a more regular trend. Overall, the experiment demonstrates that SDE simulation provides an effective numerical method for this exit problem, while also illustrating the practical influence of Monte Carlo noise and boundary-hitting discretization on the measured convergence rate.